Delivery pack FEAT-ZD-48600

Every approved agent output for FEAT-ZD-48600, in journey order

Delivery pack FEAT-ZD-48600

AP managers want to see which overdue invoices belong to vendors that usually pay late, so they can chase them first. (1) Data engineering, ap_pipeline: the vendor feed gains an optional paid_date column (ISO date, empty when unpaid); an invalid paid_date rejects the row with reason 'paid_date'; add delays_by_vendor(records) returning {vendor: [delay_days, ...]} for paid records with a paid_date, where delay_days = (paid_date - due_date) in days, in feed order. (2) AI / data science, ap_forecast: add vendor_risk(history) returning 'high' when predict_delay_days(history) > 30, 'medium' when it is > 10, otherwise 'low'; build_explanation_request gains an optional risk argument and, when given, the prompt ends with ' Risk level: <risk>.'. (3) Backend, ap_aging: add overdue_with_risk(invoices, as_of, risk_by_vendor) returning, for unpaid invoices more than 30 days past due, a list of {'id', 'vendor', 'days', 'amount', 'risk'} sorted by days descending then id, with risk 'unknown' for vendors not in risk_by_vendor. (4) Frontend, ap_portal: add renderOverdueRisk(rows) rendering '<table class="overdue-risk">' with one '<tr class="risk-RISK">' per row showing id, vendor, days and amount to two decimals, and an empty string when there are no rows. The whole flow must be proven end to end: feed -> delays -> risk -> overdue list -> portal table.

REQ-01 Requirement Intake (Requirement agent): feat.intake

feat.intake: FEAT-ZD-48600

The original intake item and its source link. Step REQ-01 (Requirement Intake), attempt 3.

Source record

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Channel	Customer Feedback	record:intake
Source	Zendesk ticket ZD-48600 (sample, synthetic)	record:intake
Received	2026-10-24	record:intake
Source link (GitHub)	`hobasa-agentic-sdlc/requirements/ZD-48600.md`	record:intake
Trigger type	Email/Ticket ("Reads the ticket")	record:brief

Original text (verbatim, treated as data)

> AP managers want to see which overdue invoices belong to vendors that usually pay late, so they can chase them first. (1) Data engineering, ap_pipeline: the vendor feed gains an optional paid_date column (ISO date, empty when unpaid); an invalid paid_date rejects the row with reason 'paid_date'; add delays_by_vendor(records) returning {vendor: [delay_days, ...]} for paid records with a paid_date, where delay_days = (paid_date - due_date) in days, in feed order. (2) AI / data science, ap_forecast: add vendor_risk(history) returning 'high' when predict_delay_days(history) > 30, 'medium' when it is > 10, otherwise 'low'; build_explanation_request gains an optional risk argument and, when given, the prompt ends with ' Risk level: <risk>.'. (3) Backend, ap_aging: add overdue_with_risk(invoices, as_of, risk_by_vendor) returning, for unpaid invoices more than 30 days past due, a list of {'id', 'vendor', 'days', 'amount', 'risk'} sorted by days descending then id, with risk 'unknown' for vendors not in risk_by_vendor. (4) Frontend, ap_portal: add renderOverdueRisk(rows) rendering '<table class="overdue-risk">' with one '<tr class="risk-RISK">' per row showing id, vendor, days and amount to two decimals, and an empty string when there are no rows. The whole flow must be proven end to end: feed -> delays -> risk -> overdue list ->

portal table.

Source: record:intake

Change requests extracted from the ticket

#	Area / repository	Change request	Source
CR-1	Data engineering / ap_pipeline	The vendor feed gains an optional `paid_date` column (ISO date, empty when unpaid). An invalid `paid_date` rejects the row with reason `'paid_date'`.	record:intake
CR-2	Data engineering / ap_pipeline	Add `delays_by_vendor(records)`. It returns `{vendor: [delay_days, ...]}` for paid records with a `paid_date`, where `delay_days = (paid_date - due_date)` in days, in feed order.	record:intake
CR-3	AI / data science / ap_forecast	Add `vendor_risk(history)`. It returns `'high'` when `predict_delay_days(history) > 30`, `'medium'` when the value is `> 10`, and `'low'` otherwise.	record:intake
CR-4	AI / data science / ap_forecast	`build_explanation_request` gains an optional `risk` argument. When `risk` is given, the prompt ends with `' Risk level: <risk>.'`.	record:intake
CR-5	Backend / ap_aging	Add `overdue_with_risk(invoices, as_of, risk_by_vendor)`. For unpaid invoices more than 30 days past due, it returns a list of `{'id', 'vendor', 'days', 'amount', 'risk'}` sorted by days descending, then by id. Risk is `'unknown'` for vendors not in `risk_by_vendor`.	record:intake
CR-6	Frontend / ap_portal	Add `renderOverdueRisk(rows)`. It renders `<table class="overdue-risk">` with one `<tr class="risk-RISK">` per row, showing id, vendor, days and amount to two decimals. It returns an empty string when there are no rows.	record:intake
CR-7	Cross-cutting	The whole flow must be proven end to end: feed -> delays -> risk -> overdue list -> portal table.	record:intake

Business need (as stated)

AP managers want to see which overdue invoices belong to vendors that usually pay late, so they can chase those first (record:intake).

Notes for the reviewer

- This file holds the original item only. The roadmap link, the checks, the assumptions and the open questions are in `feat.prebacklog.md`.
- The ticket says it is a synthetic sample (record:intake).

REQ-01 Requirement Intake (Requirement agent): feat.prebacklog

feat.prebacklog: FEAT-ZD-48600

The Pre Backlog requirement entry for this item, linked to the roadmap item and the trace ID. Step REQ-01 (Requirement Intake), attempt 3. Status: **entry created for review** (record:brief).

Entry

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Title	Overdue invoices with vendor late-payment risk, so AP managers can chase the riskiest first	record:intake
Intake item	`feat.intake.md` (Zendesk ticket ZD-48600, sample, synthetic)	record:intake
Source link (GitHub)	`hobasa-agentic-sdlc/requirements/ZD-48600.md`	record:intake
Channel	Customer Feedback	record:intake
Received	2026-10-24	record:intake
Requested by (persona)	AP managers	record:intake
Linked roadmap item	**R-AP-12**: give AP managers earlier visibility of overdue payables (Q4 2026 roadmap, AP module)	org.roadmap
Roadmap owner	PM	org.roadmap
Other roadmap item checked	R-AP-14, aging export for finance close. Not linked, because the ticket asks for no export.	org.roadmap; record:intake
Entry status	Created for review	record:brief

Affected components and planned work

Repository	Change in this item	Source
ap_pipeline	Optional `paid_date` column in the vendor feed. A row with an invalid `paid_date` is rejected with reason `'paid_date'`. New `delays_by_vendor(records)`.	record:intake
ap_forecast	New `vendor_risk(history)`, a rule on the predicted delay. Optional `risk` argument on `build_explanation_request`.	record:intake; org.model_dependency_map
ap_aging	New `overdue_with_risk(invoices, as_of, risk_by_vendor)`.	record:intake
ap_portal	New `renderOverdueRisk(rows)` table.	record:intake

Planned work: the item matches R-AP-12 on the Q4 2026 roadmap (org.roadmap).

Data flow: feed -> delays -> risk -> overdue list -> portal table. The ticket asks for this whole flow to be proven end to end (record:intake).

Checks

Decision log (conflict check)

Decision	Bearing on this item	Result	Source
D-2026-031: aging buckets use calendar days after due date	`overdue_with_risk` in ap_aging selects unpaid invoices "more than 30 days past due". Counting those days as calendar days after the due date keeps it aligned with this decision (assumption A-1). The decision covers aging buckets only, so it does not settle the day unit for `delay_days` in ap_pipeline (see OQ-6).	Consistent, provided A-1 holds	org.decision_log; record:intake
D-2026-044: monetary amounts use Decimal, never float	`overdue_with_risk` returns `amount`, and the portal shows it to two decimals. The amount should stay Decimal and be formatted only for display.	Consistent, provided Decimal is kept	org.decision_log; record:memory; record:intake
D-2026-050: paid invoices are excluded from aging	`overdue_with_risk` returns unpaid invoices only. `delays_by_vendor` uses paid records only to build payment-delay history, not the aging list.	Consistent	org.decision_log; record:intake

No conflict with a prior decision was found.

Connector inventory

- The inventory states: "No source-system connector is affected by the ap_aging library" (org.connector_inventory).
- This item also changes the **vendor feed** in ap_pipeline by adding the optional `paid_date` column (record:intake). The inventory does not say which source-system connector supplies the vendor feed, so that impact cannot be assessed here (see OQ-1).

Regulatory sources

- The source list states: "No regulatory source applies to this change" (org.regulatory_sources).
- So no section or page is captured, and no rule is affected or new.

Model dependency map

- ap_forecast uses provider acme-llm, model acme-text-2, for the payment-delay explanation (config `src/ap_forecast/models.json`) (org.model_dependency_map).
- **Affected feature: the payment-delay explanation.** When `risk` is given, the prompt that `build_explanation_request` sends to acme-text-2 ends with `' Risk level: <risk>.'` (record:intake; org.model_dependency_map). The provider and model do not change.
- `vendor_risk` uses no LLM. It is a rule on the predicted delay (org.model_dependency_map).

Shared memory (lessons to carry forward)

- L-007: test the boundary day explicitly for any day-count rule (record:memory). This item has three thresholds: more than 30 days past due in ap_aging, and a predicted delay > 30 and > 10 in ap_forecast (record:intake).
- L-011: keep Decimal arithmetic and never convert amounts to float (record:memory).

Pre Backlog sheet and DevOps board

- The workbook action asks for a check against the Pre Backlog sheet and the DevOps board (record:brief). Neither was in this step's context, so the duplicate check could not be done (see OQ-2).

Assumptions (non-blocking)

#	Assumption	Basis
A-1	Days past due in `overdue_with_risk` are calendar days from `due_date` to `as_of`. An invoice exactly 30 days past due is **not** included, because the ticket says "more than 30".	org.decision_log (D-2026-031, aging buckets); record:intake; record:memory (L-007)
A-2	`delay_days` may be zero or negative when an invoice was paid on or before its due date, and such values are kept in the history.	record:intake (the formula has no floor)
A-3	The risk thresholds are strict: a predicted delay of exactly 30 is `'medium'`, and exactly 10 is `'low'`.	record:intake ("> 30", "> 10")
A-4	The `id` tie-break in the overdue list is ascending.	record:intake ("sorted by days descending then id")
A-5	The row class `risk-RISK` takes the row's risk value, including `risk-unknown` for a vendor not in `risk_by_vendor`.	record:intake
A-6	Amounts stay Decimal and are formatted to two decimals only in the portal.	org.decision_log (D-2026-044); record:memory (L-011)

Open questions (non-blocking)

- **OQ-1:** Which source-system connector supplies the ap_pipeline vendor feed, and does it already provide a paid date? org.connector_inventory speaks only to the ap_aging library.
- **OQ-2:** The Pre Backlog sheet and the DevOps board were not in this step's context. Is there a duplicate or related open item for this request?
- **OQ-3:** What should `vendor_risk` return for a vendor with an empty payment history? The ticket does not say.
- **OQ-4:** Please confirm assumptions A-1 to A-4: the 30-day boundary, negative delays, strict risk thresholds and the ascending `id` tie-break.
- **OQ-5:** Which rounding rule applies to amounts shown to two decimals in the portal? Should vendor names be HTML-escaped in `renderOverdueRisk`?
- **OQ-6:** Should `delay_days` in `delays_by_vendor` count calendar days? The ticket says only "in days", and D-2026-031 covers aging buckets, not payment-delay history.

Notification

The workbook outcome for this step includes "Notification sent" (record:brief). This agent has not sent a notification itself. This entry is the content of that notification.